SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* (BL2, BL3)

QUESTIONS: 1 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

I M.SHARMILA (192524395) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

M. Sharmila
Signature of the student:

92
TOU

MULTI-LEVEL CONCEPT MAP → OS structures → system calls → OS functions

Operating system Types

- **Batch OS**
  - Uses job queues
  - Non-interactive
- **Time sharing OS**
  - Multi-user, multi-tasking
  - Interactive
- **Distributed OS**
  - Multiple computers
  - Resource sharing
- **Real-Time OS**
  - Time constraints
  - Predictable response

requires specific structure

OS structures

- **Monolithic**
  - single large kernel
  - simple, fast
- **Layered**
  - Modular Layers
  - Easy maintenance
- **Microkernal**
  - Minimal kernel
  - Services in user space
- **Hybrid**
  - Combines features
  - Balanced performance

determines how system calls are invoked

system calls

- Interface b/w user and Kernel
- Example: fork(), open(), read(), write()
- Used to request OS services

structure affects implementation

- **cross-link**
  - Monolithik → direct calls
  - Layered → wrapper calls
  - Microkernel → message passing
  - Hybrid → mixed approach

invoke and enable

OS functions

- Process Management (scheduling, synchronization)
- Memory Management (allocation, paging, protection)
- File system Management (files, directories, access)
- I/O Management (device drivers, buffering)
- Security & Protection (authentication, Permissions)
- User interface (CLI / GUI system interaction)

functions are achieved through system calls

CONCEPT MAP : Process management, Memory Management, File system and I/O system

